

A Kolmogorov fourth-moment bound on Poisson chaos via a martingale core

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Abstract

For any finite family of Poisson multiple integrals and any finite $p \geq 2$, we construct a common increasing filtration generated by finitely many exact Poisson counts such that the associated conditional expectations converge in L^p , remain in their original chaoses, and have bounded step kernels with finite-measure support. This finite-count martingale core allows regular fixed-chaos identities and estimates to be extended under the sole assumption of a finite fourth moment. In particular, if F lives in a Poisson chaos with unit variance and finite fourth moment, we prove that the Kolmogorov distance between F and a standard normal is bounded by $15.6\sqrt{\mathbb{E}[F^4] - 3}$. This removes Assumptions **A** and **A^{loc}** from the Kolmogorov bound of Döbler and Peccati (*Ann. Probab.*, 2018). We also obtain quantitative L^4 estimates for all iterated Malliavin derivatives and, for F in a Poisson chaos, the fourth moment assumption of F forces the L^4 -integrability of its kernel.

Keywords: Poisson chaos ; fourth moment theorem ; Kolmogorov distance ; martingale core.

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1 Introduction

This paper is motivated by the problem of establishing a Kolmogorov bound for the fourth moment theorem on the Poisson space [DP18a, DVZ18] under the sole assumption of a finite fourth moment. We achieve this goal through an intrinsic martingale approximation (Theorem 1.7) and our Kolmogorov bound (Theorem 1.9) is of the same order as the one in [DP18a] and the Wasserstein bound in [DVZ18].

1.1 Fourth moment theorems on the Poisson space

The fourth moment phenomenon was first discovered in the Gaussian setting by David Nualart and Giovanni Peccati in their groundbreaking paper [NP05]. Their result asserts that, for a sequence of normalized multiple Wiener-Itô integrals of fixed order over a Gaussian field, convergence in distribution to a standard normal $N \sim \mathcal{N}(0, 1)$ is equivalent to convergence of the corresponding fourth moments to $\mathbb{E}[N^4] = 3$. Since its introduction, the fourth moment theorem has become a widely used tool in stochastic

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analysis, especially when combined with its multivariate extension [PT05], for establishing central limit theorems for functionals of Gaussian fields. Another major methodological breakthrough was achieved by Nourdin and Peccati in [NP09], where Stein's method for normal approximation was combined with Malliavin calculus for the first time in this context. Among many other consequences, their approach yields Berry-Esseen-type bounds for a normalized multiple integral F of the form

$$\text{dist}(F, N) \leq C \sqrt{\kappa_4(F)}, \quad \text{where } \kappa_4(F) := \mathbb{E}[F^4] - 3\mathbb{E}[F^2]^2 \text{ and } N \sim \mathcal{N}(0, 1), \quad (1.1)$$

Here, dist may denote several standard probability metrics, including the total variation distance, the Kolmogorov distance (1.3), the Wasserstein distance (1.2), and the Fortet-Mourier distance. The resulting Malliavin-Stein method was subsequently developed systematically in the monograph [NP12]. See [Zhe18, Ch. 1] for a brief historical account.

We now turn to the Poisson setting, where the corresponding problems are substantially more delicate. In this framework, the Malliavin derivative D^+ is essentially an add-one difference operator rather than a derivation, and consequently no diffusion-type chain rule is available. The first systematic Poisson Malliavin-Stein estimates were obtained in [PSTU10, PZ10]. The search for a general fourth-moment theorem on the Poisson chaos remained unresolved until Döbler and Peccati [DP18a] settled the problem by combining the discrete Malliavin-Stein method, Mecke-type identities, and Ledoux's spectral approach [Led12]. We collect some important earlier results in the following.

Remark 1.1. Before the general result of Döbler and Peccati [DP18a], fourth-moment theorems on the Poisson space had been established in several important but more restrictive settings. For double Poisson integrals, qualitative normal convergence criteria in terms of kernel contractions were obtained in [PT08]. Related results for the second and third Poisson chaoses, under the regularity condition (1.6), were proved in [BP16]. For multiple integrals of arbitrary fixed order with nonnegative, or more generally constant-sign, kernels, quantitative normal approximation bounds were developed in [LRP13, ET14, Sch16]; in these settings, the relevant contraction conditions can be controlled by the fourth cumulant. Finally, for finite homogeneous sums in independent Poisson random variables with intensity parameters bounded away from zero, a universality principle and a fourth-moment theorem were established in [PZ14].

Throughout this paper, we fix an arbitrary σ -finite measure space $(\mathcal{Z}, \mathcal{Z}, \mu)$ and let η denote a Poisson random measure on $\mathcal{Z}$ with control measure μ , defined on a suitable probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mathbb{C}_q^\eta$ denote the q -th Poisson chaos associated to η , D^+ denote the add-one cost operator, $L_s^2(\mu^q)$ denote the space of symmetric functions in $L^2(\mathcal{Z}^q, \mu^q)$ and $I_q(f)$ denote a generic element of $\mathbb{C}_q^\eta$. We write $\|\bullet\|_r$ for the $L^r(\Omega)$ -norm and let d_W, d_{Kol} stand for the Wasserstein, Kolmogorov distances respectively:

$$d_W(X, Y) := \sup_{\|h'\|_\infty \leq 1} |\mathbb{E}[h(X)] - \mathbb{E}[h(Y)]| \quad (1.2)$$

$$d_{\text{Kol}}(X, Y) := \sup_{t \in \mathbb{R}} |\mathbb{P}(X \leq t) - \mathbb{P}(Y \leq t)|. \quad (1.3)$$

See Section 2.1 for further notations and background.

Now let us state the main results in [DP18a].

Theorem 1.2 (Döbler–Peccati [DP18a, Theorem 1.3]). *Fix an integer $q \geq 1$ and consider $F \in \mathbb{C}_q^\eta$ with $\mathbb{E}[F^2] = 1$. Recall the notation (1.1). Then, the following statements hold.*

(i) *Assume F satisfies Assumption A (i.e., $F \in L^4(\Omega)$ and all of the random functions D^+F , FD^+F , $(D^+F)^4$, and F^3D^+F belong to $L^1(\Omega \times \mathcal{Z}, \mathbb{P} \otimes \mu)$). Then,*

$$d_W(F, N) \leq \left(\sqrt{\frac{2}{\pi}} \frac{2q-1}{2q} + \frac{\sqrt{4q-3}}{\sqrt{q}} \right) \sqrt{\kappa_4(F)} \quad \text{with } \kappa_4(F) \geq 0. \quad (1.4)$$

(ii) Assume additionally that F satisfies Assumption $\mathbf{A}^{\text{loc}}$ (i.e., there exists a set $Z_0 \in \mathcal{Z}$ with $\mu(\mathcal{Z} \setminus Z_0) = 0$ such that $D_z^+ F$ verifies Assumption $\mathbf{A}$ for every fixed $z \in Z_0$). Then,

$$\begin{aligned} d_{\text{Kol}}(F, N) &\leq \min \left\{ 15.6, 11 + 2^{\frac{3}{2}} (\|F\|_4^2 + \|F\|_4) \right\} \sqrt{\kappa_4(F)} \\ &= 15.6 \sqrt{\kappa_4(F)}. \end{aligned} \quad (1.5)$$

Observe that the multiplicative factor in (1.5) reduces to 15.6. Indeed, since $\|F\|_4 \geq \|F\|_2 = 1$, we have $11 + 2^{\frac{3}{2}} (\|F\|_4^2 + \|F\|_4) \geq 11 + 2^{\frac{5}{2}} \approx 16.6569$. Moreover, the case $\kappa_4(F) = 0$ cannot occur by [DP18a, Prop. 1.6]; see [DVZ18, Rem. 5.2] and (1.12) below.

Döbler and Peccati themselves pointed out that Assumptions $\mathbf{A}$ and $\mathbf{A}^{\text{loc}}$ appear to be artifacts of the techniques employed in [DP18a], rather than intrinsic conditions for the fourth moment phenomenon. More precisely, these assumptions were used to justify the relevant Mecke-type identities and the almost-sure representations of the Malliavin and carré-du-champ operators appearing in their proofs. Nevertheless, Theorem 1.2 provided the first general fourth moment bound in Wasserstein and Kolmogorov distances for multiple integrals on the Poisson space. As observed in [DP18a, Rem. 1.2], Assumptions $\mathbf{A}$ and $\mathbf{A}^{\text{loc}}$ are satisfied whenever

$$\left. \begin{aligned} F &= \sum_{\ell \leq M} I_\ell(f_\ell), \text{ where } M < \infty \text{ and each kernel } f_\ell \in L_s^2(\mu^\ell) \text{ is bounded} \\ &\text{with support contained in } A^\ell \text{ for some } A \in \mathcal{Z} \text{ with } \mu(A) < \infty. \end{aligned} \right\} \quad (1.6)$$

The condition (1.6) already appeared in the work [BP16] by Bourguin and Peccati. Soon after the appearance of the work [DP18a], Döbler, Vidotto, and the author [DVZ18] used an infinitesimal exchangeable-pairs construction to establish univariate and multivariate fourth-moment bounds under the sole assumption of finite fourth moments. They obtained the following Wasserstein bound with a slightly better multiplicative factor.

Theorem 1.3 (Döbler–Vidotto–Zheng [DVZ18, Theorem 1.2 and Corollary 1.3]). *Fix an integer $q \geq 1$ and consider $F \in \mathbb{C}_q^\eta \cap L^4(\Omega)$ with $\mathbb{E}[F^2] = 1$. Then, with the notation (1.1),*

$$d_W(F, N) \leq \left(\sqrt{\frac{2}{\pi}} \frac{2q-1}{2q} + \frac{2}{3} \frac{\sqrt{4q-3}}{\sqrt{q}} \right) \sqrt{\kappa_4(F)}. \quad (1.7)$$

In particular, if $F_n \in \mathbb{C}_{q_n}^\eta$ satisfy $(\mathbb{E}[F_n^2], \mathbb{E}[F_n^4]) \rightarrow (1, 3)$, then $F_n \rightarrow N$ in law.

The derivation of the bound (1.7) relies crucially on the regularity properties of the solution to the Stein equation associated with the Wasserstein distance (1.2). Consequently, the arguments developed in [DVZ18] do not directly yield a Kolmogorov bound of order $\sqrt{\kappa_4(F)}$. In general, within Stein's method, obtaining bounds in Kolmogorov distance is more difficult than obtaining Wasserstein bounds. This is mainly because the Kolmogorov distance involves discontinuous indicator test functions, whereas the Wasserstein distance is based on Lipschitz test functions and therefore benefits from better regularity of the associated Stein solutions; see, e.g., [CGS11]. As a result, Kolmogorov bounds often require additional smoothing or concentration arguments and may incur a loss in the convergence rate. As already pointed out in [DVZ18, Rem. 1.4], the standard smoothing argument leads only to a rate of order $\kappa_4(F)^{1/4}$.

1.2 Main result: a martingale core

As already mentioned below Theorem 1.2, if a random variable $F \in \mathbb{C}_q^\eta$ satisfies the condition (1.6), Assumptions $\mathbf{A}$ and $\mathbf{A}^{\text{loc}}$ hold. In this regular case, the Kolmogorov bound (1.5) holds. It is then very natural to seek a regular approximation that converges in $L^4(\Omega)$ so that relevant fourth moment estimates can be passed along this regular approximation. Martingale convergence provides precisely this topology. However, an arbitrary Doob martingale does not preserve the Poisson chaos decomposition.

Example 1.4. Let η be a Poisson random measure on $[0, 1]$ with Lebesgue control measure, and consider $F := \eta([0, 1]) - 1 \in \mathbb{C}_1^\eta$. For $n \geq 1$, let $A_{n,k} := (\frac{k-1}{2^n}, \frac{k}{2^n}]$, $1 \leq k \leq 2^n$, and introduce the occupancy variables and a filtration:

$$\xi_{n,k} := \mathbf{1}_{\{\eta(A_{n,k}) \geq 1\}} \quad \text{and} \quad \mathcal{H}_n := \sigma\{\xi_{n,k} : 1 \leq k \leq 2^n\}, \quad n \geq 1.$$

Since the dyadic partitions are nested, $(\mathcal{H}_n)_{n \geq 1}$ is an increasing filtration. Moreover, the Poisson process is almost surely simple ([LP18, Proposition 6.9]), and hence $\sigma(\bigcup_{n \geq 1} \mathcal{H}_n) = \sigma\{\eta\}$ up to null sets. Consequently, we have the following martingale approximation: for every finite $p \geq 1$,

$$F_n := \mathbb{E}[F | \mathcal{H}_n] \xrightarrow[n \rightarrow +\infty]{\text{in } L^p(\Omega)} F.$$

However, with $\eta([0, 1]) = \sum_{k=1}^{2^n} \eta(A_{n,k})$, we can write

$$F_n = -1 + \sum_{k=1}^{2^n} \mathbb{E}[\eta(A_{n,k}) | \eta(A_{n,k}) \geq 1] \mathbf{1}_{\{\eta(A_{n,k}) \geq 1\}} = \frac{a_n}{p_n} \sum_{k=1}^{2^n} (\xi_{n,k} - p_n),$$

where $a_n := 2^{-n}$, $p_n := 1 - e^{-a_n}$, and we used the elementary fact that $\mathbb{E}[X | X \geq 1] = \frac{\lambda}{1 - e^{-\lambda}}$ for $X \sim \text{Poisson}(\lambda)$. For $z \in A_{n,k}$ and $z_1, z_2 \in A_{n,k}$, one has

$$D_z^+ F_n = \frac{a_n}{p_n} \mathbf{1}_{\{\eta(A_{n,k})=0\}} \quad \text{and} \quad D_{z_1}^+ D_{z_2}^+ F_n = -\frac{a_n}{p_n} \mathbf{1}_{\{\eta(A_{n,k})=0\}}.$$

Hence, Stroock's formula (2.4) shows that F_n does not belong to a single Poisson chaos.

Here is another naive martingale approximation by spatial truncations.

Example 1.5. Let $(Z_n)_{n \geq 1}$ be an increasing sequence in $\mathcal{Z}$ such that $\mu(Z_n) < \infty$ and $Z_n \uparrow \mathcal{Z}$. Set $\mathcal{F}_n^{\text{sp}} := \sigma\{\eta(B) : B \in \mathcal{Z}, B \subseteq Z_n\}$. For $F = I_q(f) \in L^4(\Omega)$, independence of the restrictions of the Poisson measure gives $\tilde{F}_n := \mathbb{E}[F | \mathcal{F}_n^{\text{sp}}] = I_q(f \mathbf{1}_{Z_n^q})$. Spatial conditioning does preserve the chaos order, but it does not regularize an unbounded kernel. Direct step-kernel approximation has the opposite defect: it gives regularity and L^2 -convergence, but not the L^4 -convergence supplied by a martingale.

The problem is therefore to construct a filtration whose conditional expectations simultaneously preserve the chaos order, regularize the kernels, and converge in $L^4(\Omega)$. This is precisely the purpose of the finite-count martingale core introduced below.

Definition 1.6 (Finite-partition core). For an integer $q \geq 1$, let $\mathbb{C}_q^\star$ consist of all $F \in \mathbb{C}_q^\eta$ with the property (1.6). We also put $\mathbb{C}_0^\star = \mathbb{C}_0^\eta$ as a convention.

In view of the product formula [Sur84, DP18b], every element of $\mathbb{C}_q^\star$ has moments of all orders and satisfies Assumptions **A** and **A^{loc}** of [DP18a]. The next result is the main approximation theorem of the paper.

Theorem 1.7 (Simultaneous fixed-chaos martingale core). Fix $p \in [2, \infty)$ and let

$$F_j = I_{q_j}(f_j) \in L^p(\Omega), \quad j = 1, \dots, d,$$

where $d \geq 1$, $q_j \geq 1$, and $f_j \in L_s^2(\mu^{q_j})$. There exist increasing finite-dimensional subspaces $H_n \subset L^2(\mu)$, their orthogonal projections P_n , and increasing σ -algebras

$$\mathcal{G}_n = \sigma\{\eta(A_{n,1}), \dots, \eta(A_{n,m_n})\}, \quad (1.8)$$

where the sets $\{A_{n,k} : 1 \leq k \leq m_n\}$ are pairwise disjoint with $\mu(A_{n,k}) \in (0, \infty)$ such that

$$F_{j,n} := \mathbb{E}[F_j | \mathcal{G}_n] = I_{q_j}(P_n^{\otimes q_j} f_j) \in \mathbb{C}_{q_j}^\star. \quad (1.9)$$

Moreover, for any $j \in \{1, \dots, d\}$, we have

$$F_{j,n} \xrightarrow[n \rightarrow +\infty]{\text{in } L^p(\Omega)} F_j. \quad (1.10)$$

In Example 1.4, the occupancy-based approximation mixes different Poisson chaoses, whereas in Example 1.5, the naive spatial truncation fails to satisfy the regularity condition (1.6). By contrast, Theorem 1.7 shows that conditioning on exact cell counts both preserves the chaos order and yields the desired regularity. We present the proof of Theorem 1.7 in Section 2.2.

1.3 Applications

Now let us present a few consequences of Theorem 1.7. The first application of the martingale core is a hierarchy of L^4 estimates for iterated Malliavin derivatives.

Theorem 1.8. *Let $F \in \mathbb{C}_q^\eta \cap L^4(\Omega)$ for some $q \geq 1$, and let $D^r F$ denote the r -th add-one cost derivative as in (2.3). Then, we have $D^r F \in L^4(\Omega \times \mathcal{Z}^r)$ for every $1 \leq r \leq q$ with*

$$\int_{\mathcal{Z}^r} \mathbb{E}[|D_{z_1, \dots, z_r}^r F|^4] \mu(dz_1) \cdots \mu(dz_r) \leq a_{q,r} \kappa_4(F), \quad (1.11)$$

where $a_{q,r} := \prod_{j=0}^{r-1} (4(q-j) - 3)$. In particular,

$$(q!)^4 \|f\|_{L^4(\mu^q)}^4 \leq \left(\prod_{j=1}^q (4j - 3) \right) \kappa_4(F). \quad (1.12)$$

Thus, fourth integrability of a Poisson multiple integral forces fourth integrability of its kernel. The regular core is in fact dense for the graph norm containing the L^4 norms of all derivatives up to order q ; see Corollary 2.1.

The second application of Theorem 1.7 is the anticipated Kolmogorov bound for the fourth moment theorem under the sole assumption of finite fourth moment.

Theorem 1.9 (Kolmogorov bound). *Fix an integer $q \geq 1$ and consider $F \in \mathbb{C}_q^\eta \cap L^4(\Omega)$ with $\mathbb{E}[F^2] = 1$. Then, with $N \sim \mathcal{N}(0, 1)$, $d_{\text{Kol}}(F, N) \leq 15.6 \sqrt{\mathbb{E}[F^4] - 3}$.*

Theorem 1.9 removes Assumptions **A** and **A^{loc}** from (1.5). The proof is given in Section 2.4. We also extend [DP18a, Lemma 3.2] under the sole assumption $F \in L^4(\Omega) \cap \mathbb{C}_q^\eta$, although this extension is not needed for Theorem 1.9. The finite-chaos inclusion for FG also follows from [CP25, Corollary 5.3] under the weaker assumption $FG \in L^2(\Omega)$. The additional point here is the constructive spectral closure: a common finite-count filtration yields simultaneous approximation of (F, G) , their product, and the associated carré-du-champ.

Proposition 1.10 (Graph closure of the fixed-chaos estimates). *Let $p, q \geq 1$, let $F \in L^4(\Omega) \cap \mathbb{C}_p^\eta$ and $G \in L^4(\Omega) \cap \mathbb{C}_q^\eta$. Let F_n, G_n be the martingale approximants from Theorem 1.7. Then $FG \in \bigoplus_{r=0}^{p+q} \mathbb{C}_r^\eta$. Let $\Gamma(F, G)$ denote the carré-du-champ operator (2.7). Then, we have*

$$\Gamma(F_n, G_n) \xrightarrow[\text{in } L^2(\Omega)]{n \rightarrow +\infty} \Gamma(F, G). \quad (1.13)$$

Moreover, the following exact identity holds:

$$\frac{1}{2p} \int_{\mathcal{Z}} \mathbb{E}[|D_z F|^4] \mu(dz) = \frac{3}{p} \mathbb{E}[F^2 \Gamma(F, F)] - \mathbb{E}[F^4]. \quad (1.14)$$

As a result, the estimates in [DP18a, Lemma 3.2] are valid under $F \in L^4(\Omega) \cap \mathbb{C}_p^\eta$.

2 Proofs

2.1 Basic stochastic analysis on the Poisson space

Recall that η is a proper Poisson random measure on $(\mathcal{Z}, \mathcal{Z})$ with σ -finite control measure μ ; see [LP18, Corollary 3.7]. For $q \geq 1$, let I_q denote the q th multiple Wiener-Itô integral with respect to the compensated measure $\eta - \mu$, and let $L_s^2(\mu^q)$ be the

symmetric subspace of $L^2(\mu^q)$. We use the conventions $I_0(c) = c$ and $L_s^2(\mu^0) = \mathbb{R}$. The q th Poisson chaos is

$$\mathbb{C}_q^\eta := \{I_q(f) : f \in L_s^2(\mu^q)\}, \quad \mathbb{C}_0^\eta = \mathbb{R}. \quad (2.1)$$

Every $F \in L^2(\Omega, \sigma\{\eta\}, \mathbb{P})$ admits the unique Wiener–Itô–Poisson expansion

$$F = \mathbb{E}[F] + \sum_{q \geq 1} I_q(f_q), \quad f_q \in L_s^2(\mu^q), \quad (2.2)$$

with convergence in $L^2(\Omega)$; see [LP18, Theorem 18.10].

If $F = \mathfrak{f}(\eta)$, its add-one cost is

$$D_z^+ F := \mathfrak{f}(\eta + \delta_z) - \mathfrak{f}(\eta).$$

Higher-order differences are defined recursively by $D^{(1)} = D^+$ and

$$D_{z_1, \dots, z_r}^{(r)} F := D_{z_1}^+ (D_{z_2, \dots, z_r}^{(r-1)} F), \quad r \geq 2. \quad (2.3)$$

For $F \in L^2(\Omega, \sigma\{\eta\}, \mathbb{P})$, Stroock’s formula gives

$$f_q(z_1, \dots, z_q) = \frac{1}{q!} \mathbb{E}[D_{z_1, \dots, z_q}^{(q)} F] \quad (2.4)$$

for μ^q -almost every $(z_1, \dots, z_q) \in \mathcal{Z}^q$.

The modified Wiener–Itô isometry reads

$$\mathbb{E}[I_p(f)I_q(g)] = \mathbf{1}_{\{p=q\}} q! \langle f, g \rangle_{L^2(\mu^q)}. \quad (2.5)$$

Consequently, the map

$$(f_q)_{q \geq 0} \mapsto \sum_{q \geq 0} I_q(f_q) \quad (2.6)$$

is unitary from the symmetric Poisson Fock space

$$\mathfrak{F}_\mu := \left\{ (f_q)_{q \geq 0} : \sum_{q \geq 0} q! \|f_q\|_{L^2(\mu^q)}^2 < \infty \right\}$$

onto $L^2(\Omega, \sigma\{\eta\}, \mathbb{P})$; see [LP18, Chapter 18] and [LP11, Las16].

For F with expansion (2.2), set

$$\text{dom}(D) := \left\{ F : \sum_{q \geq 1} q q! \|f_q\|_{L^2(\mu^q)}^2 < \infty \right\}.$$

For $F \in \text{dom}(D)$,

$$D_z F := \sum_{q \geq 1} q I_{q-1}(f_q(z, \cdot)),$$

with convergence in $L^2(\Omega \times \mathcal{Z}, \mathbb{P} \otimes \mu)$, and $D_z F = D_z^+ F$ for $\mathbb{P} \otimes \mu$ -almost every (ω, z) ; see [DP18a, (19)]. The Ornstein–Uhlenbeck generator is defined on

$$\text{dom}(L) := \left\{ F : \sum_{q \geq 1} q^2 q! \|f_q\|_{L^2(\mu^q)}^2 < \infty \right\}$$

by

$$LF := - \sum_{q \geq 1} q I_q(f_q).$$

Finally, whenever F, G , and FG belong to $\text{dom}(L)$, define

$$\Gamma(F, G) := \frac{1}{2} (L(FG) - F LG - G LF). \quad (2.7)$$

If $F \in \mathbb{C}_p^\eta$ and $G \in \mathbb{C}_q^\eta$, then

$$\Gamma(F, G) = \frac{1}{2} (L(FG) + (p + q)FG). \quad (2.8)$$

We refer to [Las16], [LP18, Chapter 20], and [DP18a, Section 2] for further details.

2.2 Construction of the martingale core

We are now ready to prove Theorem 1.7.

Proof of Theorem 1.7. For each j , finite linear combinations of indicators of measurable rectangles

$$B_1 \times \cdots \times B_{q_j} \quad \text{with} \quad \mu(B_i) < \infty,$$

are dense in $L^2(\mu^{q_j})$. Since symmetrization is an L^2 -contraction and f_j is symmetric, we may therefore choose symmetric rectangular step functions $u_{j,n}$ such that

$$\|u_{j,n} - f_j\|_{L^2(\mu^{q_j})} \leq 1/n. \quad (2.9)$$

Let $\mathcal{B}_n$ denote the finite collection of all coordinate sets appearing in the rectangular representations of $u_{j,k}$, for $1 \leq j \leq d$ and $1 \leq k \leq n$, and set

$$C_n := \bigcup_{B \in \mathcal{B}_n} B.$$

It is clear that $\mu(C_n) < \infty$. Let

$$\mathcal{P}_n = \{A_{n,1}, \dots, A_{n,m_n}\}$$

be the collection of non-null atoms of the finite algebra generated by $\mathcal{B}_n$ inside C_n , where complements are taken relative to C_n . Thus, the sets in $\mathcal{P}_n$ are pairwise disjoint, have finite positive measure, and form a partition of C_n modulo a μ -null set. Moreover, every $B \in \mathcal{B}_n$ is, up to a null set, a union of cells from $\mathcal{P}_n$.

Since $\mathcal{B}_n \subseteq \mathcal{B}_{n+1}$, one has

$$C_n \subseteq C_{n+1} \quad \text{and} \quad \mathcal{P}_{n+1} \text{ refines } \mathcal{P}_n \text{ on } C_n,$$

where the refinement is understood modulo μ -null sets.

Set

$$\mathfrak{H}_n := \text{span}\{\mathbf{1}_{A_{n,1}}, \dots, \mathbf{1}_{A_{n,m_n}}\} \subset L^2(\mu),$$

and let P_n denote the orthogonal projection from $L^2(\mu)$ onto $\mathfrak{H}_n$. The refinement property implies that $\mathfrak{H}_n \subseteq \mathfrak{H}_{n+1}$. Furthermore, since every coordinate set occurring in $u_{j,n}$ is a union of cells from $\mathcal{P}_n$, we have $u_{j,n} \in \mathfrak{H}_n^{\odot q_j}$, where $\mathfrak{H}_n^{\odot q_j}$ denotes the q_j -th symmetric tensor power of $\mathfrak{H}_n$. Now $P_n^{\otimes q_j}$ is the orthogonal projection from $L^2(\mu^{q_j})$ onto $\mathfrak{H}_n^{\otimes q_j}$. Since $u_{j,n} \in \mathfrak{H}_n^{\odot q_j} \subseteq \mathfrak{H}_n^{\otimes q_j}$, the best-approximation property of orthogonal projections, with (2.9), yields

$$\|P_n^{\otimes q_j} f_j - f_j\|_{L^2(\mu^{q_j})} \leq \|u_{j,n} - f_j\|_{L^2(\mu^{q_j})} \leq 1/n. \quad (2.10)$$

More explicitly, $P_n^{\otimes q_j} f_j$ has the representation

$$P_n^{\otimes q_j} f_j(z_1, \dots, z_{q_j}) = \sum_{r_1, \dots, r_{q_j}=1}^{m_n} c_{j,n}(r_1, \dots, r_{q_j}) \prod_{\ell=1}^{q_j} \mathbf{1}_{A_{n,r_\ell}}(z_\ell)$$

for suitable real coefficients $c_{j,n}(r_1, \dots, r_{q_j})$. Consequently, $P_n^{\otimes q_j} f_j$ is symmetric and bounded, is constant on each product cell of $\mathcal{P}_n^{q_j}$, and vanishes outside $C_n^{q_j}$.

Define $\mathcal{G}_n$ by (1.8). Since every cell of $\mathcal{P}_n$ is, modulo a null set, a union of cells from $\mathcal{P}_{n+1}$, each count $\eta(A_{n,k})$ is a finite sum of level- $(n+1)$ counts. Consequently, $\mathcal{G}_n \subseteq \mathcal{G}_{n+1}$.

For $1 \leq k \leq m_n$, set

$$N_{n,k} := \eta(A_{n,k}), \quad \lambda_{n,k} := \mu(A_{n,k}).$$

The random variables $N_{n,1}, \dots, N_{n,m_n}$ are independent, with $N_{n,k} \sim \text{Poisson}(\lambda_{n,k})$. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_{m_n}) \in \mathbb{N}_0^{m_n}$ (with $\mathbb{N}_0 = \{0, 1, \dots\}$), define

$$C_\alpha(N_n) := \prod_{k=1}^{m_n} C_{\alpha_k}(N_{n,k}; \lambda_{n,k}) \quad \text{and} \quad |\alpha| := \sum_{k=1}^{m_n} \alpha_k,$$

where $C_\ell(\cdot; \lambda)$ denotes the monic Charlier polynomial of degree ℓ (see [Pri09, Def. 6.2.7]). The tensor-product Charlier polynomials form a complete orthogonal system in $L^2(\mathcal{G}_n)$ with

$$C_\alpha(N_n) = I_{|\alpha|} \left(\bigotimes_{k=1}^{m_n} \mathbf{1}_{A_{n,k}}^{\otimes \alpha_k} \right).$$

See, e.g., [Pri09, Prop. 6.2.9 and Lem. 6.2.10]. The kernels on the right-hand side, with $|\alpha| = r$, span $\mathfrak{H}_n^{\otimes r}$. Grouping the complete Charlier system according to total degree therefore gives the orthogonal decomposition

$$L^2(\mathcal{G}_n) = \bigoplus_{r=0}^{\infty} I_r(\mathfrak{H}_n^{\otimes r}). \quad (2.11)$$

Equivalently, under the Poisson Fock-space isometry (2.6), the one-particle projection P_n has multiplicative second quantization $\text{Exp}(P_n) := \bigoplus_{r=0}^{\infty} P_n^{\otimes r}$. Its range is the Fock space over $\mathfrak{H}_n$, which is exactly the space in (2.11); see, e.g., [Sur84, Section 5]. Hence, $\text{Exp}(P_n)$ is the orthogonal projection onto $L^2(\mathcal{G}_n)$ and therefore

$$\text{Exp}(P_n)[X] = \mathbb{E}[X | \mathcal{G}_n] \quad \text{for} \quad X \in L^2(\Omega, \sigma\{\eta\}, \mathbb{P}). \quad (2.12)$$

Restricting (2.12) to the q_j -th chaos gives (1.9).

The martingale convergence theorem yields

$$F_{j,n} \xrightarrow[n \rightarrow +\infty]{\text{in } L^p(\Omega)} \mathbb{E}[F_j | \mathcal{G}_\infty] \quad \text{with} \quad \mathcal{G}_\infty := \sigma\left(\bigcup_{n \geq 1} \mathcal{G}_n\right).$$

On the other hand, we deduce from (2.10) that

$$\mathbb{E}[(F_{j,n} - F_j)^2] = q_j! \|P_n^{\otimes q_j} f_j - f_j\|_{L^2(\mu^{q_j})}^2 \xrightarrow{n \rightarrow +\infty} 0,$$

which ensures $\mathbb{E}[F_j | \mathcal{G}_\infty] = F_j$ almost surely. This proves (1.10) for each $j \in \{1, \dots, d\}$. $\square$

2.3 L^4 -Malliavin regularity

For $F = I_q(f)$ with $f \in L_s^2(\mu^q)$, the iterated Malliavin derivative is

$$D_{z_1, \dots, z_r}^r F = (q)_r I_{q-r}(f(z_1, \dots, z_r, \cdot)) \quad \text{with } (q)_r := \frac{q!}{(q-r)!} \quad (2.13)$$

for μ^r -almost every $(z_1, \dots, z_r) \in \mathcal{Z}^r$, $1 \leq r \leq q$. In particular, $D^q F = q!f$ is deterministic.

Proof of Theorem 1.8. Let $F_n = I_q(f_n)$ be the approximations from Theorem 1.7 with $p = 4$ such that $F_n \in \mathbb{C}_q^\star$ as in Definition 1.6. We recall from [DP18a, Lem. 3.2] that for $G \in \mathbb{C}_q^\star$ (so that G satisfies Assumption A therein)

$$\int_{\mathcal{Z}} \mathbb{E}[|D_z G|^4] \mu(dz) \leq (4q-3)\kappa_4(G). \quad (2.14)$$

Applying (2.14) to F_n gives

$$\int_{\mathcal{Z}} \mathbb{E}[|D_z F_n|^4] \mu(dz) \leq (4q-3)\kappa_4(F_n). \quad (2.15)$$

Fix $2 \leq r \leq q$. For μ^{r-1} -almost every $(z_1, \dots, z_{r-1}) \in \mathcal{Z}^{r-1}$, the random variable

$$D_{z_1, \dots, z_{r-1}}^{r-1} F_n \in \mathbb{C}_{q-r+1}^\star,$$

since its kernel is still bounded, stepwise constant, and supported in a finite-measure rectangle. Therefore, we have

$$\begin{aligned} \int_{\mathcal{Z}} \mathbb{E}[|D_y D_{z_1, \dots, z_{r-1}}^{r-1} F_n|^4] \mu(dy) &\leq (4(q-r+1)-3)\kappa_4(D_{z_1, \dots, z_{r-1}}^{r-1} F_n) \\ &\leq (4(q-r+1)-3)\mathbb{E}[|D_{z_1, \dots, z_{r-1}}^{r-1} F_n|^4]. \end{aligned}$$

Integrating in $z_1, \dots, z_{r-1}$ gives the recursion

$$\begin{aligned} M_r(F_n) &:= \int_{\mathcal{Z}^r} \mathbb{E}[|D_{z_1, \dots, z_r}^r F_n|^4] \mu(dz_1) \dots \mu(dz_r) \\ &\leq (4(q-r+1)-3) \int_{\mathcal{Z}^{r-1}} \mathbb{E}[|D_{z_1, \dots, z_{r-1}}^{r-1} F_n|^4] \mu(dz_1) \dots \mu(dz_{r-1}). \end{aligned} \quad (2.16)$$

Combining the above recursion with (2.15) yields

$$M_r(F_n) \leq a_{q,r} \kappa_4(F_n). \quad (2.17)$$

Since $F_n \rightarrow F$ in $L^4(\Omega)$,

$$\kappa_4(F_n) \rightarrow \kappa_4(F). \quad (2.18)$$

The relations (2.5) and (2.13) imply $\mathbb{E}[\|D^r(F_n - F)\|_{L^2(\mu^r)}^2] = (q)_r \mathbb{E}[(F_n - F)^2] \rightarrow 0$, so that we can select a subsequence along which $D^r F_n \rightarrow D^r F$ almost everywhere on $\Omega \times \mathcal{Z}^r$. Hence, the bound (1.11) follows from Fatou's lemma, (2.17), and (2.18). $\square$

The approximation is therefore much stronger than $L^4(\Omega)$ convergence of the random variables alone.

Corollary 2.1 (Fourth-order graph-norm density). *In the setting of Theorem 1.8, the approximants F_n can be chosen such that*

$$\mathbb{E}[|F_n - F|^4] + \sum_{r=1}^q \mathbb{E}[\|D^r(F_n - F)\|_{L^4(\mu^r)}^4] \rightarrow 0. \quad (2.19)$$

Proof. Set $R_n := F_n - F$. Then $R_n \in \mathbb{C}_q^\eta$ and $R_n \rightarrow 0$ in $L^4(\Omega)$. Using (2.16), we deduce from Theorem 1.8 that $M_r(R_n) \leq a_{q,r} \kappa_4(R_n) \leq a_{q,r} \mathbb{E}[|R_n|^4]$, which proves (2.19). $\square$

2.4 Kolmogorov bound for fourth moment theorem

Let us first record an elementary lemma.

Lemma 2.2 (Lower semicontinuity of Kolmogorov distance). *Let X_n , X , and Y be real-valued random variables. If $X_n \rightarrow X$ in distribution and the distribution function of Y is continuous, then $d_{\text{Kol}}(X, Y) \leq \liminf_{n \rightarrow \infty} d_{\text{Kol}}(X_n, Y)$.*

Proof. Choose a subsequence (n_k) along which $d_{\text{Kol}}(X_{n_k}, Y) \rightarrow \liminf_{n \rightarrow \infty} d_{\text{Kol}}(X_n, Y)$. By Skorohod representation theorem, this subsequence and X may be realized on a common probability space so that $X_{n_k} \rightarrow X$ almost surely. Let F_Y be the distribution function of Y and define $\omega_Y(\varepsilon) := \sup\{(F_Y(x + \varepsilon) - F_Y(x - \varepsilon)) : x \in \mathbb{R}\}$. A continuous distribution function is uniformly continuous on $\mathbb{R}$, because it has finite limits at $-\infty$ and $+\infty$. Hence $\omega_Y(\varepsilon) \rightarrow 0$ as $\varepsilon \downarrow 0$.

For every $x \in \mathbb{R}$ and $\varepsilon > 0$, $\mathbb{P}(X \leq x) \leq \mathbb{P}(X_{n_k} \leq x + \varepsilon) + \mathbb{P}(|X_{n_k} - X| > \varepsilon)$ and $\mathbb{P}(X_{n_k} \leq x - \varepsilon) \leq \mathbb{P}(X \leq x) + \mathbb{P}(|X_{n_k} - X| > \varepsilon)$. Consequently,

$$|\mathbb{P}(X \leq x) - \mathbb{P}(Y \leq x)| \leq d_{\text{Kol}}(X_{n_k}, Y) + \omega_Y(\varepsilon) + \mathbb{P}(|X_{n_k} - X| > \varepsilon).$$

Taking the supremum over x , letting $k \rightarrow \infty$, and then $\varepsilon \downarrow 0$, concludes the proof. $\square$

Proof of Theorem 1.9. Let F_n be the approximations of F from Theorem 1.7 with $p = 4$. Note that $\sigma_n := \sqrt{\mathbb{E}[F_n^2]} > 0$ for sufficiently large n , since $\sigma_n \rightarrow 1$.

As $F_n \in \mathbb{C}_q^\star$, the bound (1.5) holds and $d_{\text{Kol}}(F_n, \sigma_n N) = d_{\text{Kol}}(\frac{F_n}{\sigma_n}, N) \leq 15.6 \sqrt{\kappa_4(\frac{F_n}{\sigma_n})}$. Moreover,

$$\kappa_4(F_n/\sigma_n) = \frac{\kappa_4(F_n)}{\sigma_n^4} \longrightarrow \kappa_4(F), \text{ and } d_{\text{Kol}}(\sigma_n N, N) \rightarrow 0$$

since $\sigma_n \rightarrow 1$. It follows from the triangle inequality for d_{Kol} and Theorem 1.7 that

$$\liminf_{n \rightarrow +\infty} d_{\text{Kol}}(F_n, N) \leq \liminf_{n \rightarrow +\infty} [d_{\text{Kol}}(F_n, \sigma_n N) + d_{\text{Kol}}(\sigma_n N, N)] \leq 15.6 \sqrt{\kappa_4(F)}.$$

Hence, the proof is completed by invoking Lemma 2.2. $\square$

Finally, let us prove Proposition 1.10.

Proof of Proposition 1.10. F_n and G_n are homogeneous Charlier polynomials of total degree p and q , respectively. Consequently,

$$F_n G_n \in \bigoplus_{r=0}^{p+q} \mathbb{C}_r^\star.$$

From the $L^4(\Omega)$ -convergence $(F_n, G_n) \rightarrow (F, G)$, one has $F_n G_n \rightarrow FG$ in $L^2(\Omega)$. The space $\bigoplus_{r=0}^{p+q} \mathbb{C}_r^\eta$ is a closed subspace of $L^2(\Omega)$, which proves $FG \in \bigoplus_{r=0}^{p+q} \mathbb{C}_r^\eta$.

By definition, the Ornstein-Uhlenbeck generator L is bounded on $\bigoplus_{r=0}^{p+q} \mathbb{C}_r^\eta$. Therefore,

$$L(F_n G_n) \longrightarrow L(FG) \text{ in } L^2(\Omega),$$

and (1.13) follows from the spectral definition of the carré-du-champ; see (2.7)–(2.8).

Corollary 2.1 gives $DF_n \rightarrow DF$ in $L^4(\Omega \times \mathcal{Z})$, so that

$$\int_{\mathcal{Z}} \mathbb{E}[|D_z F_n|^4] \mu(dz) \longrightarrow \int_{\mathcal{Z}} \mathbb{E}[|D_z F|^4] \mu(dz). \quad (2.20)$$

For every n , [DP18a, Lemma 3.2] gives

$$\frac{1}{2p} \int_{\mathcal{Z}} \mathbb{E}[|D_z F_n|^4] \mu(dz) = \frac{3}{p} \mathbb{E}[F_n^2 \Gamma(F_n, F_n)] - \mathbb{E}[F_n^4].$$

Note that the relation (1.13) and $F_n \rightarrow F$ in $L^4(\Omega)$ imply the $L^1(\Omega)$ -convergence of $F_n^2 \Gamma(F_n, F_n) \rightarrow F^2 \Gamma(F, F)$, which together with (2.20) proves (1.14). $\square$

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